FIITJEE

ALL INDIA TEST SERIES

FULL TEST – XVI

JEE (Main)-2021

TEST DATE: 08-08-2021

ANSWERS, HINTS & SOLUTIONS

Physics

PART – A

SECTION – A

1. A

Sol. $mg = V\rho g$

$= 3 \times 2 \times (0.01\ m) \times (10^3\ kg/m^3) \times 10$

$Mg = 600$

$\Rightarrow m = 60\ kg$

2. B

Sol. $V_Q - V_O = \frac{1}{2}B(\omega)(r^2) = \frac{B\omega}{2}(2a)^2 = 2B\omega a^2.$

3. D

Sol. For electron in electric field, magnitude of its momentum may increase, decrease or remain constant w.r.t. initial value.

4. B

Sol. $0.6 = \frac{Q_1 - Q_2}{Q_1} = 1 - \frac{20}{Q_1}$

$\Rightarrow Q_1 = 50, W = Q_1 - Q_2 = 30\ J$

5. D

Sol. Here wave velocity is $\frac{\omega}{K}=\frac{100\pi}{5\pi}$ = 20 m/s

So $T=\mu v^2=0.1\times(20)^2$ = 40 N

6. C

Sol. Variation of moment of inertia with temperature

$I = I_o (1 + 2\alpha \Delta T)$

C.O.A.M:

$I_i\omega_i = I_f \omega_f$

7. B

Sol. Total momentum = $\int_{\theta=-90}^{+90}(dm)v\cos\theta$

$dm = \frac{M}{\pi R}\times R d\theta$

Total momentum = $\int\frac{M}{\pi R}vRd\theta\cos\theta=\frac{2Mv}{\pi}$

8. C

Sol. $V_{AC}=2\times\left(\frac{0.2}{1.8+0.2}\right)$ …(i)

Also, V_{AC} = $(1.5)\left(\frac{5}{15}\right)\left(\frac{\ell}{100}\right)$ …(ii)

9. B

Sol. $F=\frac{dp}{dt}=\left(\sqrt{2}\,V\right)\frac{dm}{dt}=\left(\sqrt{2}\,V\rho\right)\frac{d(volume)}{dt}=\left(\sqrt{2}\,V\rho\right)(AV)$

10. B

Sol. Factual based.

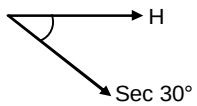
11. D

Sol. $A_1V_1 = A_2V_2 \Rightarrow a\sqrt{2gh}=A\left(-\frac{dh}{dt}\right)$

$\Rightarrow \quad \frac{a}{A}\int_0^t dt=-A\int_{h_1}^{h_2}\frac{dh}{\sqrt{2gh}}$

12. B

Sol.



$\therefore$ H sec 30° = $2H/\sqrt{3}$

13. C

Sol. Potential of centre of sphere = $\frac{Kq}{r}+V_i=\frac{Kq}{r}$

where V_i = potential due to induced charge at centre = 0 [$\therefore$ Σq_i = 0 and all induced charges are equidistance from centre]

$\therefore$ potential at point $P = \frac{Kq}{r} = \frac{Kq}{r_1} + V_i$ (For conductor all points are equipotential)

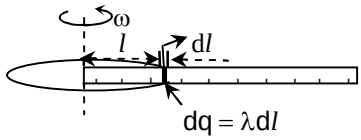
$\therefore$ $V_i = K\left(\frac{q}{r} - \frac{q}{r_1}\right)$

14. C

Sol. $\frac{dv}{dr} = -F$

15. D

Sol. We can visualize the rod to consist of differential elements dQ, which constitute a series of concentric current loops. The charge per unit length of the rod $\lambda = \frac{Q}{L}$.



So the charge on a differential element of length dl.

$dq = \lambda\, dl$

The current dI due to rotation of this charge is given by

$$dI = \frac{dq}{(2\pi/\omega)} = \frac{\omega}{2\pi} dq = \frac{\omega}{2\pi}\lambda dl$$

The magnetic moment of this differential current loop.

$$d\mu = dI(\pi l^2) = \left(\frac{\omega}{2\pi}\lambda\, dl^2\right)\pi\, l^2 = \frac{\omega\lambda}{2\pi} l^2 dl$$

Total magnetic moment is $\mu = \frac{\omega\lambda}{2}\int_0^L l^2 dl = \frac{\omega\lambda\, L^3}{6} = \frac{Q\omega\, L^2}{6}$

Again, $\left(\frac{Q}{2m}\right) l = \left(\frac{Q}{2m}\right)\left(\frac{1}{3}mL^2\right)\omega = \left(\frac{QL^2\omega}{6}\right) = \mu$

16. A

Sol. For the reflection at the concave mirror,

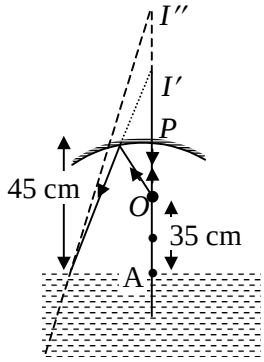
u = – 10 cm; v = ?; f = – 15 cm

From the mirror formula, we have

$$v = \frac{uf}{u-f} = \frac{(-10)\times(-15)}{-10+15} = \frac{150}{5} = +30 \text{ cm}$$

The positive sign indicates that the image is formed on the other side of the concave mirror,

Now, the image formed by the concave mirror serves as a virtual object for refraction at water surface which takes placed from air to water. So,

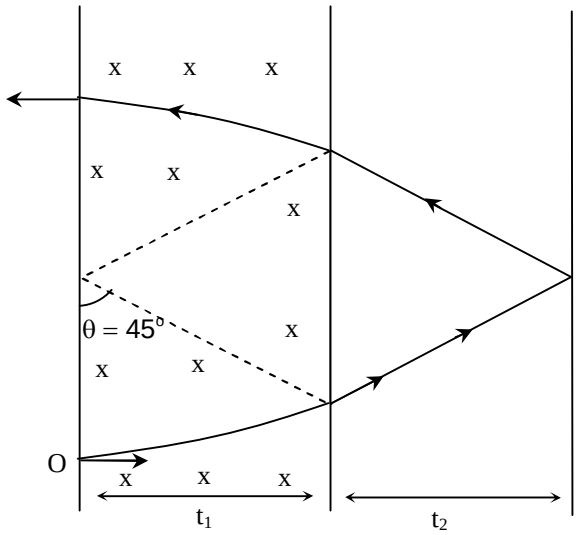


$\mu = \frac{\text{Apparent height}}{\text{Real height}}$

$\therefore$ AI" = Apparent height = μ × real height = $\frac{4}{3}\times 75$ = 100 cm

17. A

Sol.



$$t_1 = \frac{\theta}{\omega} = \frac{\frac{\pi}{4}}{\frac{qB}{m}} = \frac{m\pi}{4qB}$$

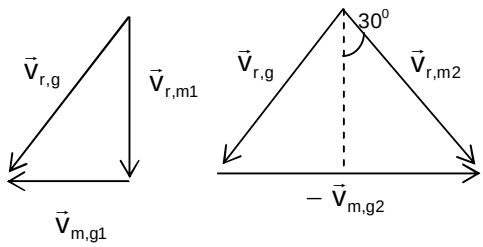
$$t_2 = \frac{\frac{mv}{\sqrt{2}qB}}{\frac{v}{\sqrt{2}}} = \frac{m}{qB}$$

$$T = 2(t_1 + t_2) = \left(\frac{m\pi}{2qB} + \frac{2m}{qB}\right)$$

$$= \frac{m}{2qB}(\pi + 4)$$

18. D

Sol.



19. B

Sol. $$f = \frac{(2n+1)v}{4(l+e)}$$

$$(l_1 + e) = \frac{v}{4f}$$

$$(l_2 + e) = \frac{3v}{4f} \Rightarrow \frac{l_2 + e}{l_1 + e} = 3$$

$l_2 = (3.6 - 2.34)$ m and $l_1 = (3.6 - 3.22)$

$\Rightarrow$ e = 0.06 m = 0.6 r $\Rightarrow r$ = 0.1 m

$A = 100\ \pi$ cm^2

20. A

Sol. $N = N_o e^{-\lambda t}$

$0.99 = 1\, e^{-\lambda t} \Rightarrow \lambda t = \ln\left(\frac{100}{99}\right)$

$50 \text{ sec} = t = \frac{\ln\left(\frac{100}{99}\right)}{\lambda}$

$\Rightarrow \quad t_{1/2} = \frac{50\ln(2)}{\ln(100/99)}$

SECTION – B

21. 00057.73

Sol. 2N = 400Newton

N = 200 Newton

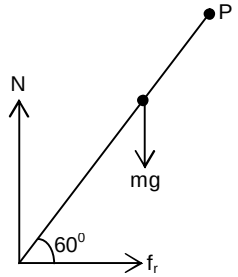
Net torque about P

$\tau_P = f_r \sin 60^0 + mg \sin 30^0 \frac{l}{2}$

$-N \sin 30^0 l = 0$

$\Rightarrow f_r \frac{\sqrt{3}}{2} l + \frac{200}{2} \times \frac{l}{2} - \frac{200}{2} \times l = 0$

$f_r = \frac{100}{\sqrt{3}}$



22. 00001.67

Sol. about ICR

$\omega \times 3 = 2 \Rightarrow \omega = \frac{2}{3}$

$V_{cm} = \omega \times \frac{5}{2} = \frac{2}{3} \times \frac{5}{2} = \frac{5}{3} m/s$

23. 00002.15

Sol. $\frac{17\lambda}{4} = l = \lambda = \frac{4l}{17}$

$V = \sqrt{\frac{Y}{\delta}} = \sqrt{\frac{2 \times 10^{11}}{7.8 \times 10^3}} = \sqrt{\frac{10^9}{39}}$

$\delta = \frac{V}{\lambda} = \frac{17}{4l}\sqrt{\frac{10^9}{39}} = 10^4$

$l = \frac{17}{4}\sqrt{\frac{10}{39}} = 2.15$

24. 00023.04

Sol. $F = \frac{\mu_0}{4\pi}\frac{q^2 v^2}{r^2} = \frac{10^{-7} \times \left(1.6 \times 10^{-19} \times 3 \times 10^5\right)^2}{\left(10^{-3}\right)^2}$

$= 9 \times 2.56 \times 10^{-29}$

25. 00002.83

Sol. $I_{eff} = \sqrt{(2)^2 + (2)^2} = 2\sqrt{2}$ & $F = il_{eff}B = 2 \times 100\sqrt{2}$

26. 00002.25

Sol. $R_x = \dfrac{R_1R_2}{R_1 + R_2} = 2.25$

27. 00001.56

Sol. kinetic energy of portion of length $\lambda = \dfrac{1}{2} \times 0.01 \times a^2 \times 200 \times \pi^2 \times \lambda$

$= a^2\pi^2\lambda$

28. 00002.25

Sol. $\Delta l = \dfrac{2Fl}{AY} + \dfrac{3Fl}{AY} + \dfrac{4Fl}{AY} = \dfrac{9Fl}{AY}$

29. 00001.56

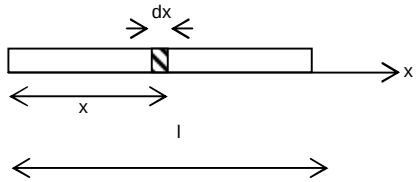
Sol. $I = \dfrac{P}{4\pi r^2} = \dfrac{0.8}{4\pi \times 2.25} = \dfrac{0.8}{9\pi}$

$I = 2\pi^2\delta V^2A^2f^2 = \dfrac{0.8}{9\pi}$

$A = \dfrac{3.12}{2} \times 10^{-7}$

30. 00002.75

Sol. $X_{cm} = \dfrac{\int_0^l ax^2dx}{\int_0^l axdr}$



$= \dfrac{2l}{3}$

$mass = m = \int_0^l axdx = \dfrac{al^2}{2}$

moment of inertia about 0

$I = \int_0^l axx^2dx = \int_0^l ax^3dx = \dfrac{al^4}{4}$

$I_{cm} + mx_{cm}^2 = I$

$I_{cm} = I - mx_{cm}^2 = \dfrac{al^4}{9} - \dfrac{al^2}{2} \times \dfrac{4l^2}{9}$

$= al^4\left\{\dfrac{1}{4} - \dfrac{2}{9}\right\} = \dfrac{al^4}{36}$

Chemistry

PART – B

SECTION – A

31. C

Sol. One mole of NH_3 occupy 22.4 L at NTP, not at 300 K and 1 atm.

32. D

Sol. Sulphur forms two sigma bonds with two lone pairs.

33. B

Sol. $(CH_3)_2 SiCl_2 \xrightarrow{H_2O} (CH_3)_2 Si(OH)_2 \xrightarrow{-H_2O}$ Silicones

34. C

Sol. HCOOH is stronger than CH_3COOH. Hence the solution of HCOOH will contain more no. of ions than that of CH_3COOH.

35. B

Sol. $\frac{\Lambda_m}{\Lambda_e} = 3 \Rightarrow \Lambda_m = 3\Lambda_e$

n-factor of $MgCl_2.KCl.6\ H_2O = 3$.

36. B

Sol. $NaOH \xrightarrow{CrO_2Cl_2} Na_2CrO_4 + H_2O + Cl^-$

$Pb^{2+} + 2Cl^- \longrightarrow PbCl_2 \downarrow$

$Pb^{2+} + CrO_4^{2-} \longrightarrow PbCrO_4 \downarrow$

37. B

Sol. One Na^+ is removed and remaining three(effective no. per unit cell) are present.

38. B

Sol. In (B) the NH_2 group which is at a larger distance from the COOH group, should be more basic it should accept H^+ ion from acid.

39. A

Sol. $2LiOH \xrightarrow{\Delta} Li_2O + H_2O$

40. C

Sol. $CH_3CHO \xrightarrow[KCN]{NH_4Cl} MeCH(NH_2)CN \xrightarrow[H_2O]{HCl} MeCH(NH_2)COOH$

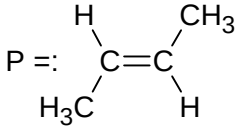
41. B

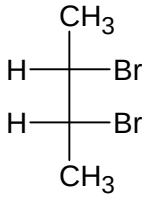
Sol. $KO_2 \xrightarrow{H_2O} KOH + H_2O_2 + O_2$

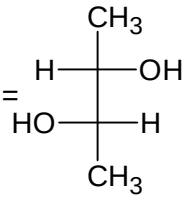
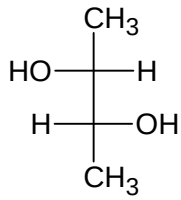
$KH + H_2O \longrightarrow KOH + H_2$

42. D

Sol.

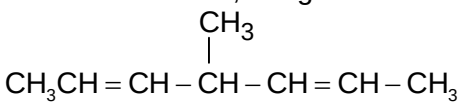
P =: 

Q = 

R or S =  or 

43. C

Sol. Due to double bond, the geometrical isomers are:



Cis	Cis
Trans	Trans
Cis	Trans
Trans	Cis

Identical

When the double bonds are Cis and trans, the molecule becomes optically active.

$\therefore$ For trans Cis (two optical isomers are possible)

$\therefore$ Total no. of stereoisomers are:

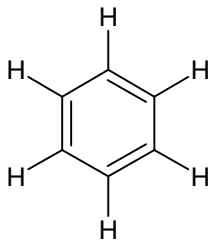
Geometrical(2) + Optical(2) = 4

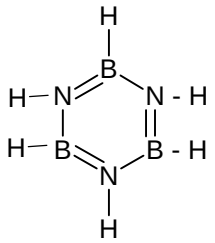
44. B

Sol. $CO_3^{2-} + H^+ \longrightarrow HCO_3^-$ (Weak base does not dissociate much). Therefore, the reaction proceeds to forward direction by removing CO_3^{2-} as HCO_3^-.

45. D

Sol.

Organic benzene 

Inorganic benzene 

46. B

Sol. Aniline will be deactivated in presence of H^+.

47. C

Sol. For zero order reaction,
Rate = K
Both rate and rate constant are independent of concentration.

48. D

Sol. pH of $NaH_2PO_4 = \frac{p^{K_{a_1}} + p^{K_{a_2}}}{2} = \frac{7.2 + 10.6}{2} = 8.9$

49. D

Sol. Oxidation of side chain takes place.

50. A

Sol. SCN^- ion is formed in sodium extract of organic compounds containing C, N and S elements.

$$Fe^{3+} + 3SCN^- \longrightarrow \underset{\text{Red colour}}{Fe(SCN)_3}$$

SECTION – B

51. 00000.10

Sol. Na_2CO_3 does not react with $KMnO_4/H^+$
M_{eq} of $KMnO_4/H^+ = 400 \times 0.5 \times 5 = 1000$
$\therefore M_{eq}$ of $Na_2C_2O_4 = 1000$
or $\frac{W}{E} \times 1000 = 1000$
$\therefore \frac{W}{E} = 1$
n-factor of $Na_2C_2O_4$ is 2 as it is converted to CO_2
$\therefore$ Moles of $Na_2C_2O_4 = = \frac{\text{Equivalent}}{\text{n} - \text{factor}} = \frac{W/E}{2} = \frac{1}{2}$
$\therefore$ Mole fraction $= \frac{1/2}{5} = 0.1$

52. 00150.00

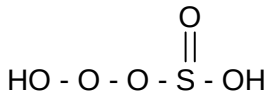
Sol. $C_{rms} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3(120)}{16 \times 10^{-3}}} = 150 \text{ ms}^{-1}$

53. 00055.80

Sol. There is a large jump from $I.E_3$ which means the atom contains two valence electron.
$\therefore$ It's valency is 2
Let the atom is M
So formula of it's oxide = MO
Molar mass = 39.8 + 16 = 55.8 g mol^{-1}

54. 00002.00

Sol. The structure of H_2SO_5 is:



55. 00004.00

Sol. $\Delta T_f = K_f \times n \times \frac{1000}{W}$

Or, $4 = 1.86 \times \frac{6.2}{62} \times \frac{1000}{W}$

$\therefore$ W = 46.5 g

$\therefore$ Mass of water converted to ice = 50.5 – 46.5 = 4 g

56. 00003.00

Sol. Only NH_2 group absorb CH_3I. Since it is primary amine, three moles can be absorbed to form quaternary salt.

57. 00058.00

Sol. β-keto acids undergo decarboxylation on heating forming ketones

58. 00005.00

Sol. X is H_3PO_4

59. 00006.00

Sol. $pH = p^{K_a} + \log\frac{[CH_3COONa]}{[CH_3COOH]} = 5 + \log\frac{10}{1} = 6$

60. 00008.00

Sol. It contains three asymmetric carbon atoms

$\therefore$ Number of optical isomers = $2^3 = 8$

Mathematics

PART – C

SECTION – A

61. A

Sol. $\frac{f(1+x)-f(1)}{x}=1+g(x)\Rightarrow f'(1)=1$

Also $f(xy)=f(x)f(y)$

$\Rightarrow \frac{y}{x}=\frac{f'(x)}{f(x)}.\frac{f(y)}{f'(y)}$ put y=1

$\Rightarrow \frac{f(x)}{f'(x)}=x\Rightarrow \int \frac{f(x)}{f'(x)}dx=\frac{x^2}{2}+c$

62. A

Sol. $0<b<1, \log_b x$ is decreasing in $0<\alpha<\frac{\pi}{4}$ then $0<\sin\alpha<\cos\alpha<1$

$\Rightarrow \log_b \sin\alpha>\log_b\cos\alpha>0$

and $(\sin\alpha)^{\log_b\sin\alpha}<(\sin\alpha)^{\log_b\cos\alpha}<(\cos\alpha)^{\log_b\cos\alpha}$

63. C

Sol. $f(t)=\int_0^t f(t-y)g(y)dy$, put $t-y=x$ and $g(y)=y$

$=\int_0^t f(x)(t-x)dx$, we have $f(x)=e^x$

$=t\int_0^t e^x dx-\int_0^t xe^x dx$

$=e^t-1-t$

64. D

Sol. We get p=0, q=$3\pi/4$, r=π

65. B

Sol. There are 3 vertical lines, 3 horizontal lines 3 each of slopes 1 and -1, and 2 each with slopes $2,-2,\frac{1}{2}$ and $-\frac{1}{2}$ for total 20 lines

66. D

Sol. Let the coordinates of P be (α,β)

Then $PQ=2\beta$ and $OP=\sqrt{\alpha^2+\beta^2}$

Since OPQ is an equilateral triangle OP = PQ

$\Rightarrow \quad \alpha^2+\beta^1=4\beta^2\Rightarrow\alpha^2=3\beta^2$

$\Rightarrow \quad \alpha=\pm\sqrt{3}\beta$

Also since (α,β) lies on the given hyperbola, $\frac{\alpha^2}{a^2}-\frac{\beta^2}{b^2}=1$

$$\Rightarrow \frac{3\beta^2}{a^2}-\frac{\beta^2}{b^2}=1\Rightarrow \frac{1}{b^2}=\frac{1}{\beta^2}>0\Rightarrow \frac{b^2}{a^2}>\frac{1}{3}.$$

$$\Rightarrow\ e^2-1>\frac{1}{3}\Rightarrow e^2>\frac{4}{3}\Rightarrow e>\frac{2}{\sqrt{3}}$$

67. D

Sol. put $4+\frac{1}{x^2}=t$

68. C

Sol. Given $b^2=ac,\ x=\frac{a+b}{2},\ y=\frac{b+c}{2}$

Now $\frac{a}{x}+\frac{c}{y}=\frac{2a}{a+b}+\frac{2c}{b+c}=\frac{2a}{a(1+r)}+\frac{2ar^2}{ar(1+r)}=2$

And $\frac{b}{x}+\frac{b}{y}=\frac{2b}{a+b}+\frac{2b}{b+c}=2\left[\frac{ar}{a+ar}+\frac{ar}{ar(1+r)}\right]=2$

69. C

Sol. $\frac{b+c}{b-c}=\frac{2+\sqrt{3}}{\sqrt{2}-1}=\frac{\tan\frac{B+C}{2}}{\tan\frac{B-C}{2}}=\frac{2+\sqrt{3}}{\tan\left(\frac{B-C}{2}\right)}$ $\left[\text{using } \tan\frac{B-C}{2}=\frac{b-c}{b+c}\cot\frac{A}{2}\right]$

$\therefore$ $B-C=45^o$. But $B+C=150^o$

$\therefore$ $C=52\frac{1}{2}^o$

70. B

Sol. g(x) is defined if $f(x+1)$ is defined. Thus the domain is all $x\,|\,x+1\in[0,2]\rightarrow x\in[-1,1]$.
Since $f(x+1)\in[0,1],-f(x+1)\in[-1,0]$. Thus $g(x)=1-f(x+1)\in[0,1]$ is the range of g(x).
Thus the answer is $[-1,1],[0,1]$

71. B

Sol. 2nd equation has imaginary roots $\Rightarrow$ both roots are identical

$\Rightarrow$ $a=1,b=\sqrt{2},\ c=1;\ \cos C=\frac{1+2-1}{(2)(\sqrt{2})}=\frac{1}{\sqrt{2}}\ \Rightarrow\ \angle C=45^o$

72. A

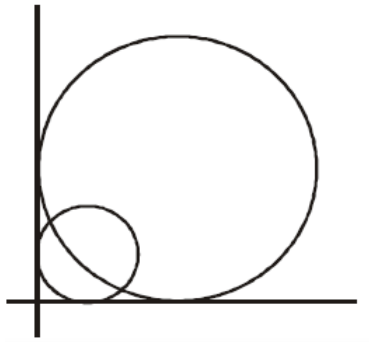
Sol. Circle is $(x-r)^2+(y-r)^2=r^2$

$x^2+y^2-2xr-2yr+r^2=0$

Hence the circles are

$x^2+y^2-2xr_1-2yr_1+r_1^2=0$ ....(1)

$x^2+y^2-2xr_2-2yr_2+r_2^2=0$ ....(2)



(1) and (2) are orthogonal hence

$2r_1r_2+2r_1r_2=r_1^2+r_2^2$

$4\frac{r_1}{r_2}=\left(\frac{r_1}{r_2}\right)^2=1\Rightarrow\left(\frac{r_1}{r_2}\right)^2-4\left(\frac{r_1}{r_2}\right)^2+1=0$

$\Rightarrow\frac{r_1}{r_2}=\frac{4\pm\sqrt{16-4}}{2}=\frac{2\pm2\sqrt{3}}{2}$

$=2+\sqrt{3}$ or $2-\sqrt{3}$

73. D

Sol. The position vector of points D, E, F are respectively $\frac{i+j}{2}+k$, $i+\frac{k+j}{2}$ and $\frac{i+k}{2}+j$

So, position vector of centroid of $\Delta DEF=\frac{1}{3}\left[\frac{i+j}{2}+k+i+\frac{k+j}{2}+\frac{i+k}{2}+j\right]$

$=\frac{2}{3}[i+j+k]$,

74. C

Sol. $(x-a)$ is factor of $f'(x), f''(x), \ldots\ldots f^{(2n)}(x)$

$\Rightarrow f(x)=b\pm(x-a)^{2n+2}$ but given that f(x) has maximum value b at x=a

$\Rightarrow f(x)=b-(x-a)^{2n+2}$

75. C

Sol. $\cot\frac{B}{2}\cot\frac{C}{2}=\frac{s}{s-a}=\frac{2}{2s-2a}$

$\frac{a+b+c}{b+c-a}=\frac{4a}{2a}=2$

76. A

Sol. As given two circles touches each other. Only one common tangent exists.

77. B

Sol. We have $\log_{10}\left(ab^3\right)=1 \Rightarrow ab^3=10$ .....(1)

Also $\log_{10}\left(a^2b\right)=1 \Rightarrow a^2b=10$ .....(2)

$\therefore$ From equation (1) and (2), we get

$ab^3 = a^2b$

$\Rightarrow \quad ab\left(b^2 - a\right) = 0$

$\therefore b^2 = a\left(\text{As } a \neq 0, b \neq 0\right)$

Now on putting $a = b^2$ in equation (1), we get

$b^5 = 10$

$\therefore \quad b = 10^{\frac{1}{5}}$ and $a = 10^{\frac{2}{5}}$

Hence $\log_{10}(ab) = \log_{10}\left(10^{\frac{2}{5}}10^{\frac{1}{5}}\right) = \log_{10}\left(10^{\frac{3}{5}}\right) = \frac{3}{5}$

78. D

Sol. $a^{2010} - 1 = (a-1)\left(a^{2009} + ...1\right)$

now $a - 1 = \frac{a^{2010} - 1}{a^{2009} + ...1}$

let put $a = \sqrt[2010]{(1+x)}$ then

$\lim\limits_{x\to 0} \frac{\sqrt[2010]{1+x} - 1}{2x + x^2}$

$= \lim\limits_{a\to 1}\left(\frac{1}{2 + a^{2010} - 1}\right)\left(\frac{1}{a^{2009} + ...1}\right)$

$= \frac{1}{2}\frac{1}{2010}$

79. A

Sol. The probability of 4 being the minimum number $= \frac{{}^6C_2}{{}^{10}C_3}$

(because, after selecting 4 any two can be selected from 5, 6, 7, 8, 9, 10).

The probability of 8 being the maximum number $= \frac{{}^7C_2}{{}^{10}C_3}$.

The probability of 4 being the minimum number and 8 being the maximum number $= \frac{3}{{}^{10}C_3}$

$\therefore$ the required probability $= P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$= \frac{{}^6C_2}{{}^{10}C_2} + \frac{{}^7C_2}{{}^{10}C_3} - \frac{3}{{}^{10}C_3} = \frac{11}{40}$

80. B

Sol. Applying $\left(C_3 \to C_2 - C_1\right)$

$$D=\begin{vmatrix}3^2+k & 4^2 & 3\\ 4^2+k & 5^2 & 4\\ 5^2+k & 6^2 & 5\end{vmatrix}=0$$

$$\Rightarrow\begin{vmatrix}9+k & 16 & 3\\ 7 & 9 & 1\\ 9 & 11 & 1\end{vmatrix}=0 \quad \left(R_3\to R_3-R_2; R_2\to R_2-R_1\right)$$

$$\Rightarrow k-1=0\Rightarrow k=1$$

SECTION – B

81. 00021.00

Sol. (mean – mode) = 3 mean – median)
Hence 3 (mean – median) = 63
$\Rightarrow$ mean – median = 21

82. 00003.00

Sol. We have $\dfrac{4+\sec 20^o}{\operatorname{cosec} 20^o}=\dfrac{\sin 20^o}{\cos 20^o}\left(4\cos 20^o+1\right)$

$$=\frac{2\sin 40^o+\sin 20^o}{\cos 20^o}$$

$$=\frac{\sin 40^o+\left(\sin 40^o+\sin 20^o\right)}{\cos 20^o}$$

$$=\frac{\sin 40^o+2\sin 30^o\cos 10^o}{\cos 20^o}$$

$$=\frac{2\sin 60^o\cos 20^o}{\cos 20^o}=2\times\frac{\sqrt{3}}{2}=\sqrt{3}$$

Hence square of the value of expression = 3

83. 00001.00

Sol. Note that for the x interval $(-\infty,0), f(x)$ is strictly increasing and has range $(-\infty,\infty)$.

For the x interval $(0,\infty)$, again, $f(x)$ is strictly increasing and has range $(-\infty,\infty)$.

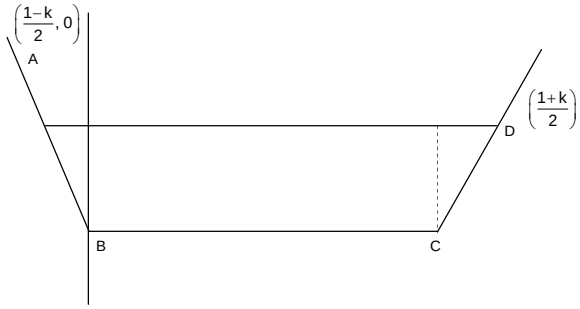
Hence for all a, there is exactly one positive and one negative solution to $f(x)=a$.

Thus, there are 2^n solutions to $f^n(x)=a$, and the answer is $2^3=8$

84. 00005.00

Sol. $-2 = 1(1-k) + 2\frac{1}{2}(1-k)\left(\frac{1+k}{2}-1\right)$

$\Rightarrow k^2 = 5$



85. 00003.00

Sol. $x^3 = t$ in I_2

$3x^2dx = dt \Rightarrow 3I_2 = \int_0^1 \frac{1}{e^t(2-t)}dt$

$= \int_0^1 \frac{1}{e^{(1-t)}(2-(1-t))}dt$

$3eI_2 = \int_0^1 \frac{e^t}{1+t}dt = I_1$

Therefore $\left|\frac{I_1}{eI_2} = 3\right|$

86. 00002.00

Sol. $\frac{z_2}{z_1} = ik$, k is real, $z_2 = ikz_1$

$\Rightarrow \frac{6z_1 - 8z_2}{4z_2 + 3z_1} = \frac{6z_1 - 8ikz_1}{4ikz_1 + 3z_1} = \frac{6 - 8ki}{4ik + 3}$

$\therefore \left|\frac{6z_1 - 8z_2}{4z_2 + 3z_1}\right| = \left|\frac{6 - 8ik}{4ik + 3}\right| = \frac{\sqrt{36 + 64k^2}}{\sqrt{16k^2 + 9}} = \sqrt{4} = 2$

87. 00004.00

Sol. 2 or 3 or 4 or 5 copies of each book out of 5 identical copies can be selected in 4 ways. Since number of books is n, therefore, total number of selections = 4^n

Given $4^n = 256 \Rightarrow n = 4$

88. 00017.00

Sol. Let $S_n = 1.2^1 + 2.2^2 + 3.2^3 + 4.2^4 + ... + n.2^n$ ……(1)

$\Rightarrow \quad 2.S_n = 1.2^2 + 2.2^3 + ... + (n-1)2^n + n.2^{n+1}$ ……(2)

$\therefore$ On Subtracting, we get

$\Rightarrow \quad -S_n = \left(2^1 + 2^2 + 2^3 + ... + 2^n\right) - n.2^{n+1}$

$= 2\left(2^n - 1\right) - n.2^{n+1}$

$\therefore\ S_n = n.2^{n+1} - 2\left(2^n - 1\right) = n.2^{n+1} - 2^{n+1} + 2 = 2 + (n-1)2^{n+1}$

But $2 + (n-1)2^{n+1} = 2 + 2^{n+5}$ (given)

$\Rightarrow \quad n - 1 = 2^4$

$\therefore \quad n = 17$

89. 00001.50

Sol. Let $b = f(a)$

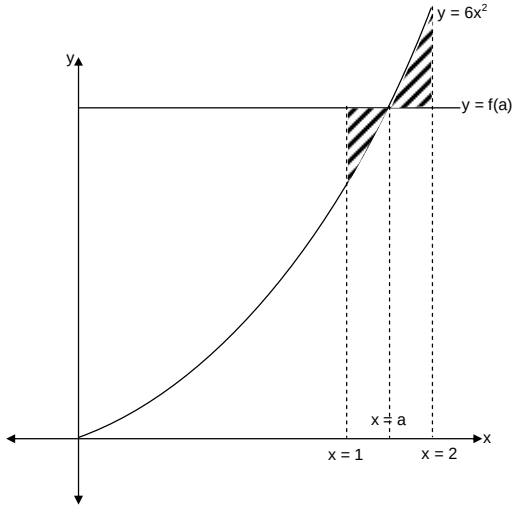
$$A = \int_1^a \left(b - 6x^2\right)dx + \int_a^2 \left(6x^2 - b\right)dx$$

$$= \left|bx - 2x^3\right|_1^a + \left|2x^3 - bx\right|_a^2$$

$$= 8a^3 - 18a^2 + 18$$

For minimum area $\frac{dA}{da} = 0$

$\Rightarrow 24a^2 - 36a = 0 \quad \Rightarrow \quad a = 1.5$



90. 00004.00

Sol. Here, sides of triangle be a = 5p, b = 6p and c = 5p

$\because \quad s = \frac{a+b+c}{2}$

$$= \frac{5p + 6p + 5p}{2} = 8p$$

$\therefore$ Radius of incircle $r = \frac{\Delta}{s} = \frac{\sqrt{s(s-a)(s-b)(s-c)}}{s}$

$\Rightarrow \quad 6 = \frac{\sqrt{(8p)(3p)(2p)(3p)}}{8p}$

$\Rightarrow \quad 6 = \frac{4p \times 3p}{8p}$

$\Rightarrow \quad p = 4$